THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 11: **Electricity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A cell of emf 6 V is connected across 2 ohm and 4 ohm in parallel. The total power delivered is 1

(a) 27 W **(b)** 6 W **(c)** 9 W **(d)** 18 W

2. The effective resistance between two ends when four 4 ohm resistors are in a parallel-series arrangement of two in series in each of two parallel branches is 1

(a) 8 ohm **(b)** 4 ohm **(c)** 16 ohm **(d)** 2 ohm

3. Assertion (A): The fuse wire is made of a metal of low melting point. Reason (R): It must melt when the current exceeds the safe limit. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. When two resistors are connected in series the equivalent is 9 ohm and in parallel it is 2 ohm. The resistors are 1

(a) 4 ohm and 5 ohm **(b)** 2 ohm and 7 ohm **(c)** 1 ohm and 8 ohm **(d)** 3 ohm and 6 ohm

5. A heater coil has resistance R and gives out heat Q in a time t. If it is cut to half its length and used on the same supply, the heat in the same time is 1

(a) Q **(b)** 2 Q **(c)** Q / 2 **(d)** 4 Q

6. Assertion (A): The equivalent resistance of two resistors in parallel is more than that of each. Reason (R): More paths are available for the flow of current. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. Two bulbs of 60 W and 100 W (both rated 220 V) are connected in series across 220 V. The bulb that glows brighter is 1

(a) 60 W **(b)** 100 W **(c)** both equally **(d)** neither

8. A battery supplies 36 W to two identical resistors connected in series. If they are connected in parallel to the same battery, the power would be 1

(a) 72 W **(b)** 144 W **(c)** 18 W **(d)** 36 W

9. **Assertion (A):** In a series combination of resistors, the current is the same through every resistor. 1
Reason (R): In a series combination the potential difference across each resistor is the same.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A fuse wire is made of a material of high melting point. 1
Reason (R): A fuse protects a circuit by melting when an excessive current flows through it.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A resistor dissipates 100 W when connected to 100 V. What power will it dissipate at 50 V? 2

12. An electric heater draws 10 A at 220 V. Find the energy used in 2 hours in kWh and the cost at Rs 6 per unit. 2

13. A 12 ohm resistor and a 4 ohm resistor are in parallel with a total current of 4 A. Find the current in the 4 ohm resistor. 2

14. Why does the filament of a bulb break more easily when switched on? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Resistors of 1 ohm, 2 ohm and 3 ohm are connected in series to a 12 V source. Find the power dissipated in each and the total. 3

16. Why do we use the kWh instead of the joule for household energy bills? 3

17. Why is the electric power transmitted at a very high voltage over long distances? 3

OR

A wire of resistance R is stretched to double its length and then folded into two equal parts connected in parallel. Find the resulting resistance. 3

18. Why does a heater coil shorten if a part of it breaks and is rejoined, and what happens to its power? 3

OR

In a circuit, two resistors of 6 ohm and 12 ohm are in parallel and the current in the 6 ohm resistor is 2 A. Find the total current and the voltage. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Priya has three resistors of $2\ \Omega$, $3\ \Omega$ and $6\ \Omega$ and a 6 V battery. She first joins the three resistors in series and then in parallel. 5
(i) Find the equivalent resistance in the series combination.
(ii) Find the equivalent resistance in the parallel combination.
(iii) Find the total current drawn from the battery in the parallel combination.
(iv) Find the current through each resistor in the parallel combination and verify that they add up to the total current.

20. Explain how the V-I characteristics are used to find the resistance when two readings are given: at 2 V the current is 0.4 A and at 6 V the current is 1.2 A. Is the conductor ohmic? Find the power at 6 V. 5

OR

The following readings are taken for a resistor: V = 1 V, I = 0.1 A; V = 2 V, I = 0.2 A; V = 3 V, I = 0.3 A. Find the resistance from the graph, the power at 3 V and verify Ohm's law. If the same resistor is a wire of area 1 mm^2 and length 5 m, find the resistivity. 5

END OF QUESTION PAPER — Total: 40 Marks